EXP NO : 1 Average Marks and Highest Average Subject

Aim: To calculate the average marks of each subject and find the subject with the highest average using NumPy.

Algorithm:

1. Import NumPy.
2. Create a 4×4 matrix of student marks.
3. Calculate subject averages using np.mean().
4. Find the highest average using np.argmax().
5. Display the results.

Program:

```
import numpy as np

student_scores = np.array([
    [80, 85, 75, 70],
    [90, 88, 80, 78],
    [70, 75, 85, 80],
    [85, 90, 90, 85]
])

subjects = ["Math", "Science", "English", "History"]

averages = np.mean(student_scores, axis=0)

print("Average marks:", averages)

highest = np.argmax(averages)
print("Highest average subject:", subjects[highest])
```

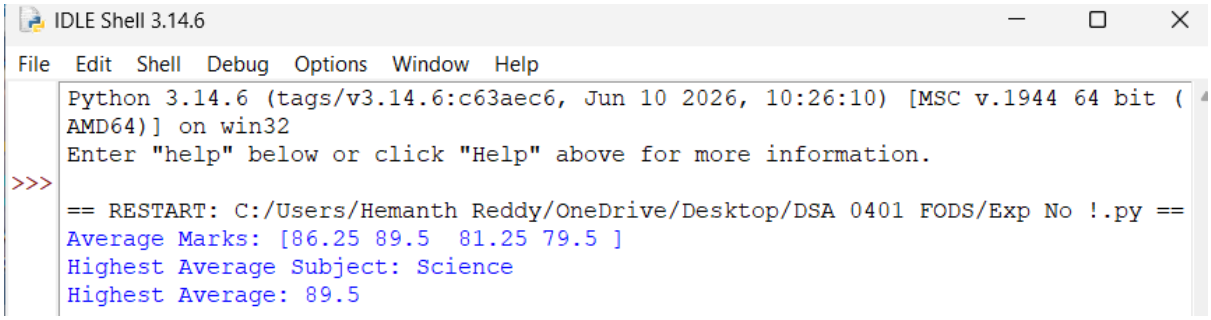
print("Highest average marks:", averages[highest])

Output:

Average marks: [81.25 84.5 82.5 78.25]

Highest average subject: Science

Highest average marks: 84.5



Result: The average marks of each subject are calculated, and Science has the highest average of 84.5 marks.